Definitions:

(i) **Item** : An item or an entity is defined as a functional or structural unit like a component, device, equipment or a system which can be considered individually and may include hardware and //or software and human resources.

(ii) **Lifetime** : The time taken by any non-repairable item in its failure from the start of its initial operation.

(iii) **Fault:** this is the state of an item when it is unable to perform its required function due to some reasons other than preventive maintenance work or some other planned actions. Fault can be a defect or failure.

(iv) **Defect** : This is like nonconformity of a requirement in an item which is related to its intended purpose. Defects occur due to some errors committed during design or production process.

(v) **System** : It is consisting of assemblies, subsystems, components along with skills and techniques which can be operated independently. All of these tangible and intangible entities interact with each other to perform a set of required functions by a system.

(vi) **Functional Block**: It may be a simple component or a subassembly of components or a large subsystem as parts of system. Any functional block is said to be failed if it is not able to perform its required function. The functions and performance criterion of these functions for each element of the system is always predefined. The potential failure modes of these functional block and their effect on the system is determined.

(vii) **Functional Block Diagram**: It is used to represent the structure of the system.

(viii) **Systematic Failure** : These failures are also called dynamic defect which are deterministic in its nature and occur due to a cause which can be removed by modifying the design or the manufacturing process or operating procedure or some other relevant factor.

(ix) **Useful Life** : This is the time interval for an item, starting from the time when it begin its operation under given conditions till the time when its frequency of failures become unbearable and it is considered, not fit for repair i.e. worthy to be replaced.

(x) **Mission Profile** : It is a specific task which has to be completed by an item by operating under given conditions in a given specified time.

When any system or equipment is introduced into the market and it reaches to the consumer from the manufacturer , it is always expected that initially the system is free from any defect and will perform its required or intended function without any failure for a stated period of time and will show a safe behavior upon any catastrophic failure. Nowadays , the concepts of reliability, maintainability, availability and safety have come into large existence due to the high cost of system failure during operation.

Lecture 24

Now , the answer of whether any system will operate without any failure under stated conditions during a specified time , can not be given as yes or no, it will always be in probability terms.

All components or items of given type , even produced by the same manufacturer, using same materials and under same operating conditions do not fail at the same time i.e. if m units of these item which are completely identical are put into operation initially , at time t=0 to perform its required function then all units will fail at different time. Hence the failure time of the item will not be same, it will a random variable and will follow probability distribution function by using which we can determine the probability of its survival within a given interval or beyond some time or before some time. The exact form of probability distribution may or may not be known and will vary as the item is produced by different manufacturer or by using the material produced by different supplier or made under different operating conditions and accordingly the probability calculation may get changed. The calculated probabilities are function of time and hence random variable which is considered for probability evaluation is Time.

Initially any component or item is considered to be good when it starts operating and then its probability of failure increases as time passes and it tends to one as time tends to infinity. The failure time or the time to failure of any item or equipment is the time interval, starting from the time when the item is put into operation until it fails for the first time. The time may be measured by indirect time like number of rotations of a bearing etc, hence it may be discrete or continuous where discrete variable can be approximated by continuous variable.

Let T be a failure time continuous random variable following the probability distribution function F(t) and the probability density function f(t) where

$$f(t) = \lim_{\Delta t \to 0} \frac{P(t \le T \le t + \Delta t)}{\Delta t} \quad \text{and} \quad F(t) = P(T \le t) \quad where\ F(0) = 0\ and\ F(\infty) = 1$$

The cumulative distribution function F(t) is also called <u>a life time distribution function</u>.

The complementary cumulative distribution function is given by

$$\bar{F}(t) = 1 - F(t) = P(T > t) \quad where\ \bar{F}(0) = 1\ and\ \bar{F}(\infty) = 0$$

This is also **<u>called the survival function or the reliability function since</u>** it gives the **<u>probability of survival of the item up to a particular time t and is denoted by R(t) hence</u>**

$$R(t) = \int_t^{\infty} f(v)dv = 1 - \int_0^t f(v)dv$$

Hence Reliability is the characteristic of an item or an equipment which define its ability to perform its function successfully without any interruption. It is always represented in probability terms.

Definition (Reliability): It is defined by the probability that any item or equipment will perform its specified task for which it is specifically designed , under stated conditions for stated interval of time without any failure.

The concept of Reliability can be applied to any process or service also. The failure of an item occurs when it is not able to perform its required function. Initially when an item starts operating, it is considered to be free from any defect or systematic failure. Failures may be classified by mode of failure which is symptom of observing the failure like cracking of components, or by cause of failure (which may be intrinsic in nature and occur due to weakness or wear out of the item or it may be extrinsic in nature which may occur due to any mistake committed during designing , production or operating phase). Failure may also due to physical , chemical or any other process. Failures may be sudden and are called catastrophic failure or it may be gradual and is called degradation failure.

Reliability can be measured quantitatively for a non-repairable system where it is calculated till the time of its first failure or a repairable system

Definitions:

Fault mode and effect analysis : (FMEA) –This is the qualitative method where all possible fault modes for each element of the item are analyzed and its effects on other elements as well as on the functioning of whole system is investigated.

Fault mode , effect and critical analysis : (FMECA) : this is the extension of FMEA where quantitative approach is also used by calculating the probability of occurrences of all failure modes. On the basis of probability, the ranking is determined and possibilities are analyzed to reduce or to completely eliminate their effect , on the basis of their ranking.

Fault Tree analysis (FTA) : It is a top-down approach in the form of tree to determine which fault mode of the element and /or external event resulted in a given fault of the item i.e. a combination of potential failures and events are determined which may result to a specified system failure.

Reliability block diagram: It is network diagram represents how the functioning of various functional blocks result in the successful functioning of the system.

Failure Rate or Hazard rate : Consider an item or equipment which is not failed in time t then its conditional probability of failure in time $(t, t+\Delta t)$ is called its failure or hazard rate $\varphi(t)$ which is given by

$$\emptyset(t) = \lim_{\Delta t \to 0} \frac{P(t \le T \le t + \Delta t | T > t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{P\,(t \le T \le t + \Delta t \;\cap\; T > t)}{\Delta t \;\; P(T > t)}$$

$$= \lim_{\Delta t \to 0} \frac{P\,(t \le T \le t + \Delta t\;)}{\Delta t \;\; P(T > t)} = \frac{f(t)}{1 - P(T \le t)} = \frac{f(t)}{\bar{F}(t)} = \frac{f(t)}{R(t)} \qquad (1)$$

$$where\ f(t) = \frac{d}{dt}(F(t)) = \frac{d}{dt}\left(1 - \bar{F}(t)\right) = -\frac{d}{dt}\left(\bar{F}(t)\right) = -\frac{d}{dt}(R(t))$$

$$Hence\ \ \varphi(t) = \frac{1}{R(t)}\left[-\frac{d}{dt}(R(t))\right] = -\frac{d}{dt}(Ln\, R(t)) \qquad (2)$$

$$\textbf{Further}\ \ \ln R(t) = -\int_0^t \emptyset(v)dv \;\; => \;\; R(t) = e^{-\int_0^t \varphi(v)dv} \qquad (3)$$

$$\textbf{Hence}\ \ f(t) = \varphi(t)R(t) = \varphi(t)\, e^{-\int_0^t \varphi(v)dv} \qquad (4)$$

Failure rate function which is the function of life distribution of the item is also called the force of mortality which is different from rate of occurrence of failures of a repairable item.

Note: (1) Experimentally, the concept of hazard rate can be understood like total m number of identical components are exposed to failure or we can say they are put into operation at time t=0 and tested for their time of failure. Hazard rate is dependent on the number of failures in a given time period and the number of components exposed to failure.

We have $m_s(t) + m_f(t) = m$ *where $m_s(t)$ are the no. of components* which will be surviving till time t and others failed so that

$$R(t) = \frac{m_s(t)}{m} = 1 - \frac{m_f(t)}{m} = 1 - F(t)$$

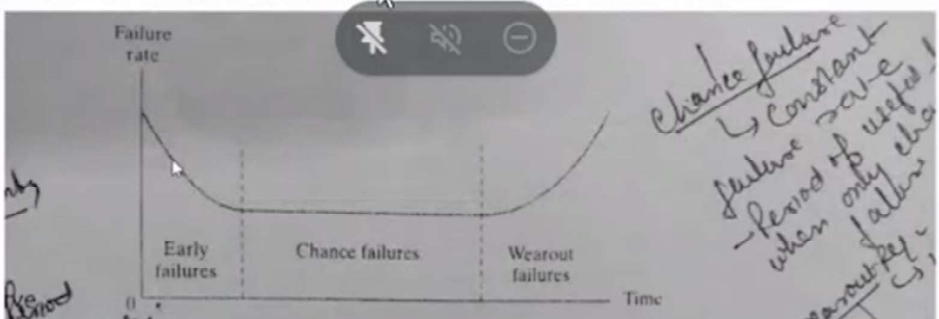
$$\frac{dR(t)}{dt} = -\frac{1}{m}\frac{dm_f(t)}{dt} \quad and\ since\ f(t) = -\frac{dR(t)}{dt} = \frac{1}{m}\frac{dm_f(t)}{dt}$$

Therefore at time t=0, when m components are exposed to failure, probability density function and hazard rate are same.

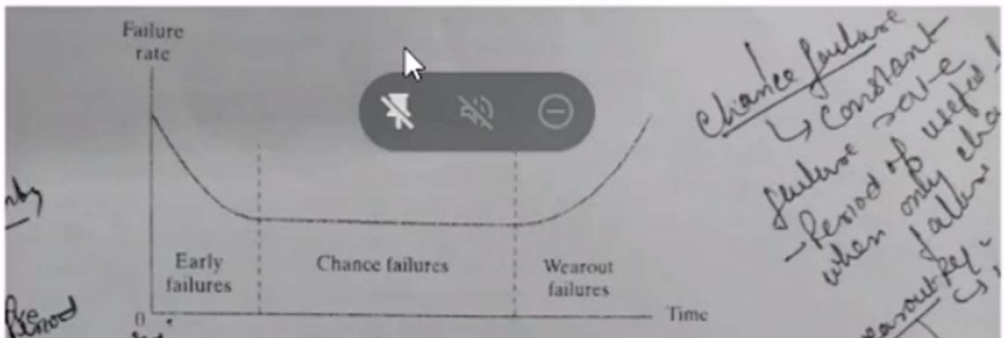
At time t, hazard rate is given by $\emptyset(t) = \frac{1}{m_s(t)}\frac{dm_f(t)}{dt} = \frac{m}{m_s(t)}\left[\frac{1}{m}\frac{dm_f(t)}{dt}\right] = \frac{1}{R(t)}f(t)$

(2) The failure rate or hazard rate function follows a bathtub curve which indicates that when the item starts operating initially, it is having a high failure rate since some hidden defects in the item become visible and result in the failure of item which is also called infant mortality. If the item is able to survive in this period , failure rate stabilizes at

certain level , remain so for some time and then it starts increasing as item ages and starts wearing out. The Bathtub shape of hazard rate divide the lifetime of item into three time period zones : the burn-in period, useful life period which is also called chance failure period and the wear-out period. Here burn-in period is the time during which item is tested at factory before its delivery to the customer or in the market.



A failure-rate curve that is typical of many manufactured items is shown in Figure 15.2. The curve is conveniently divided into three parts. The first part is characterized by a decreasing failure rate and it represents the period during which poorly manufactured items are weeded out. (It is common in the electronics industry to burn in components prior to actual use in order to eliminate any early failures.) The second part, which is often characterized by



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Mean time to failure (MTTF) : It is defined as following:

$$MTTF = E(T) = \int_0^\infty t f(t) dt \qquad (5)$$

This is also called MTBF (mean time between failure) if the time to repair or replace the item upon its failure is assumed to be negligible otherwise if it is significant then MTBF also includes MTTR(mean time to repair)

$$As\ we\ know\ \ f(t) = -\frac{dR(t)}{dt},\ \ (5)\ can\ be\ \ written\ as$$

$$MTTF = \int_0^\infty t \left[-\frac{dR(t)}{dt}\right] dt \ = \ -tR(t)|_0^\infty + \int_0^\infty R(t) dt$$

$$MTTF = \int_0^\infty R(t) dt \qquad (since\ R(\infty) = 0\ and\ \ R(0) = 1) \qquad (6)$$

Note : **(1)** MTTF can also be derived by taking the Laplace transformation of R(t) i.e. L(R(t)=R(s) and taking s=0.

(2) Since MTTF represents the mean or measure of central tendency of Life time distribution, it can also be measured by median life T_M which divides the distribution into two parts and item has 50% chance of survival before T_M i.e. $R(T_M)=0.5$.

Also it can be measured by mode which is another measure of central tendency and represents the time where the chances of failure of item is maximum i.e. life time distribution attains maximum frequency. Hence $f(T_{M0}) = \max_{0 \le t \le \infty} f(t)$.

Mean Residual life (MRF) :

The probability of the survival of an item up to time t2 if it has survived till time t1 where t2>t1 and y=t2 - t1

$$R(y|t1) = P(T > t2|T > t1) = \frac{P(T > t2)}{P(T > t1)} = \frac{R(t2)}{R(t1)}$$

i.e. the probability of the survival of the item at age t1 for y additional units.

$$MRL(t1) = \beta(t1) = \int_0^\infty R(y|t1)dy = \frac{1}{R(t1)}\int_{t1}^\infty R(y)dy$$

Example: Consider the following reliability function

$$R(t) = \frac{1}{(4t+3)^2} \quad t \geq 0 \ \ where\ t\ is\ measured\ in\ years$$

We have $f(t) = -\frac{dR(t)}{dt} = \frac{2*4}{(4t+3)^3}$ $\quad and \quad \emptyset(t) = \frac{f(t)}{R(t)} = \frac{8}{(4t+3)}$

$$MTTF = \int_0^\infty R(t)dt = \left(\frac{1}{12}\right) years$$

4

Mean residual life at the age of 2 months=2/12years=1/6 years

$$MRL\left(\frac{1}{6}\right) = \frac{1}{R(\frac{1}{6})}\int_{1/6}^\infty \frac{1}{(4y+3)^2}dy = \frac{9}{121}\frac{3}{44} \approx 0.005\ years$$

Example : If the failure time of an item follows Exponential distribution then its probability density function is given by

$$f(t) = \vartheta e^{-\vartheta t} \qquad \vartheta > 0, \qquad t > 0$$

The reliability function of the item is

$$R(t) = P(T > t) = \int_t^\infty f(y)dy = \int_t^\infty \vartheta e^{-\vartheta y}dy = e^{-\vartheta t}$$

The hazard rate is given as

$$\emptyset(t) = \frac{f(t)}{R(t)} = \frac{\vartheta e^{-\vartheta t}}{e^{-\vartheta t}} = \vartheta$$

Hence any item whose life distribution function is exponential, is having constant hazard rate.

Further $MTTF = \int_0^\infty R(t)dt = \int_0^\infty e^{-\vartheta t}dt = \frac{1}{\vartheta}$

$$and\ \ P(T > MTTF) = P\left(T > \frac{1}{\vartheta}\right) = R\left(\frac{1}{\vartheta}\right) = e^{-1}$$

$$\boldsymbol{MRL(t) = \beta(t) = \int_0^{\infty} R(y|t1)dy = \frac{1}{R(t1)}\int_{t1}^{\infty} R(y)dy = \frac{1}{e^{-\vartheta t1)}}\int_{t1}^{\infty} e^{-\vartheta y}dy = \frac{1}{\vartheta} = MTTF}$$

Example: Consider a component with constant failure rate $\varphi(t) = \vartheta = 0.00034/hour$

(i) The probability of survival of the component for 200 hours is given by

$$\boldsymbol{R(t) = e^{-\vartheta t} = e^{-0.00034*200} = e^{-0.068} \approx 0.93}$$

(ii) $\boldsymbol{MTTF = \frac{1}{\vartheta} = \frac{1}{0.00034} \approx 2941.18\ hours}$

(iii) $\boldsymbol{The\ probability\ of\ survival\ its\ MTTF = P(T > MTTF) = \frac{1}{e} \approx 0.37}$

(iv) ***The probability* of survival for 45 days where 1 day=24 hours and 45 days=1080 hours is given by**

$$\boldsymbol{R(t) = e^{-\vartheta t} = e^{-0.00034*1080} = e^{-0.3672} \approx 0.69}$$

(v) Let us assume that the component is able to survive 150 hours without failure. Then the probability of failure of the component in 180 hours is

$$\boldsymbol{P(T \leq 180|T > 150) = \frac{P(T \leq 180\ and\ T > 150)}{p(T > 150)} = \frac{F(180) - F(150)}{R(150)}}$$

$$\boldsymbol{= \frac{e^{-0.00034*150} - e^{-0.00034*180}}{e^{-0.00034*150}} = P(T \leq 30) = 1 - e^{-0.00034*30} \approx 0.01}$$